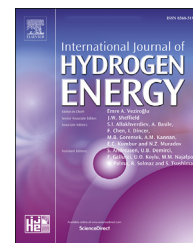


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Ag-doped TiO₂ photocatalysts with effective charge transfer for highly efficient hydrogen production through water splitting

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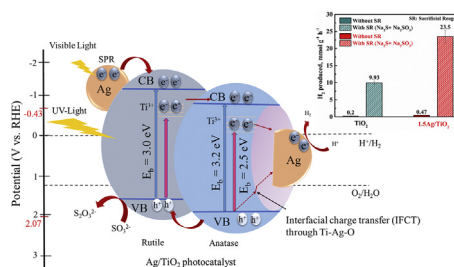
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HIGHLIGHTS

- The Ag/TiO₂ synthesized by chemical reduction, lowers the band gap of TiO₂.
- XPS confirms oxygen vacancies and Ag in both Ag⁰ and Ti–Ag–O phases.
- The Ti–Ag–O phase promotes efficient charge transfer and separation.
- Oxygen vacancies and Ti–Ag–O in Ag/TiO₂ enhances the photoreduction of water to H₂.
- Remarkable H₂ production of 23.5 mmol g^{−1} h^{−1} over Ag doped TiO₂.

GRAPHICAL ABSTRACT



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ABSTRACT

The development of efficient metal doped semiconductors for solar energy harvesting to produce hydrogen has attracted significant attention. Herein, the H₂ generation over Ag-doped TiO₂ photocatalyst, synthesized using a simple and cost-effective method based on chemical reduction, was reported. The Ag/TiO₂ exhibited an absorption peak in the visible region and the reduction of the bandgap to 2.5 eV due to surface plasmonic resonance (SPR). X-ray photoelectron spectroscopy revealed the presence of oxygen vacancies and 11% of Ag in Ti–Ag–O phase. The effect of reaction time and photocatalyst loading in the absence and presence of sacrificial reagents (alcohols and sulfur) on water splitting was studied and compared the activity of Ag/TiO₂ with that of bare TiO₂. The H₂ production rate of 23.5 mmol g^{−1} h^{−1} (with an apparent quantum yield of 19%), over 1.5Ag/TiO₂, was the highest ever reported so far. The observed higher activity could mainly be attributed to the existence of oxygen vacancies and the Ti–Ag–O phase. The photocatalyst was stable for

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three consecutive cycles in both the presence and absence of sacrificial reagents. This study offers new insights into the rational design of metal-support hybrid structures for hydrogen production through photocatalytic water splitting.

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Introduction

In recent years, the photocatalytic water splitting to produce hydrogen from solar energy is gaining interest [1–4]. TiO_2 , a well-known photocatalyst for water splitting, possesses several drawbacks such as wide bandgap energy (3.2 eV) and faster charge (e^-/h^+) recombination. However, it can be modified using various techniques, including metal and non-metal doping [5], and coupling with other semiconductors such as CdS [6], ZnO [7], to overcome these drawbacks.

Metal doping into the TiO_2 lattice induces the localized surface plasmon resonance (SPR) which can tailor the bandgap of TiO_2 and thereby enhancing the photocatalytic activity of TiO_2 [8,9]. The SPR is the collective oscillation of confined free electrons on metallic nanoparticles (NP). Due to SPR, metal-doped TiO_2 can efficiently absorb solar radiation and improve the charge transfer efficiency of the semiconductors through electron trapping [10–13]. In addition, the metal dopants have a positive effect in e^-/h^+ separation and charge transfer by inducing oxygen vacancies and by the formation of Ti^{3+} in TiO_2 lattice [14,15]. Among the noble metals, silver (Ag) is an attractive dopant due to its plasmonic properties in visible range along with extraordinary physico-chemical properties and lower cost [16].

Wu et al. (2019) studied Ag-doped TiO_2 and reported the formation of metallic Ag, Ag_2O and AgO species on the surface of TiO_2 [17]. Liu et al. (2012) reported simultaneous deposition (or impregnation) and doping of Ag/Ag(I) on TiO_2 . The formed Ag(I) contributes to the visible light absorption and the metallic Ag promotes the effective electron-hole separation [18]. Ortiz et al. (2015) reported the highest H_2 production from metallic Ag than Ag incorporated into the TiO_2 lattice [19]. Duan et al. (2017) investigated the role of Ag and oxygen vacancies in Ag– TiO_2 nanocomposites for promoting the effective e^-/h^+ separation [20]. Recently, Nanaji et al. (2019) confirmed the combined effect of doped- and metallic-Ag on TiO_2 for an enhanced photocatalytic activity [21].

In most of the above studies, the Ag-doped TiO_2 photocatalysts were synthesized using sol-gel, impregnation, and hydrothermal method followed by chemical- or photo-reduction [22,23]. For Chemical-reduction, strong reducing agents such as NaBH_4 , N_2H_4 which are not eco-friendly were used [24,25]. Anna et al. (2010) reported higher bioactivity of ascorbic acid (AA) than N_2H_4 in the reduction of Ag ions [26]. Wang et al. (2016) tested Ag/ TiO_2 synthesized by using AA as a reducing agent and reported a uniform distribution of Ag on the top and inner surface of TiO_2 [27]. The lower redox potential AA helps to obtain monodispersed, non-agglomerated Ag nanoparticles [28]. AA acts as an ideal electron donor for Ag_2O reduction to form Ag^0 nuclei on the TiO_2 surface [29].

Most of the Ag-doped TiO_2 photocatalysts were tested in the application of photocatalytic dye degradation [30], antimicrobial activity [31], disinfection activity [17], nanofiltration membrane [32], etc.

Research efforts have also been made on the usage of the sacrificial reagents (SRs) to reduce the charge carrier recombination and to improve H_2 production from the water splitting reaction. The well-known SRs include organic acids [33], alcohols [34], EDTA [35], and redox mediators ($\text{Fe}^{3+}/\text{Fe}^{2+}$, IO_3^-/I^- , etc.) [36]. Utilization of by-products from industry, such as sulfides and sulfites, as SRs is highly helpful in mitigating the waste generation.

The oxygen vacancies and Ti^{3+} centers in the TiO_2 lattice can improve the life of photogenerated charge carriers [37–39]. In addition, the synergistic effect of anatase/rutile phase TiO_2 was found to exhibit higher activity by trapping e^-/h^+ at the phase junction. Although there are reports on photodegradation using Ag loaded anatase/rutile TiO_2 heterostructures, the literature is scarce for the generation of H_2 using this photocatalyst system [40–42]. To the best of our knowledge, the synergistic effects of oxygen vacancies, surface Ti^{3+} , and doped and metallic Ag on anatase/rutile TiO_2 heterostructures for the production of H_2 are not reported in the open literature.

In this work, an environmentally benign, economical and easy technique for uniform doping of Ag on TiO_2 by chemical reduction using ascorbic acid was developed. The synthesized Ag/ TiO_2 catalysts were characterized by high-end characterization techniques such as XPS and FETEM to determine the Ag state and its distribution, respectively. The catalysts were also characterized using various spectroscopic and photoelectrochemical (PEC) techniques to determine the optical and electrical properties. The photocatalyst with enhanced properties such as reduced bandgap and mixed-phase (Ag^0 and Ti-Ag-O) was tested for its activity and stability in water splitting to produce hydrogen both in the presence and absence of sacrificial reagents. The mechanism of water splitting activity of doped semiconductor was explained based on the Ti^{3+} species, oxygen vacancies and the mixed phase of metallic Ag and Ti-Ag-O formation.

Experimental section

Materials

Silver nitrate (99%) was procured from Loba Chemie Pvt. Ltd. India. Ascorbic acid (99%), sulfuric acid (99%), Ammonium hydroxide (30%) and 2-propanol (99%) were purchased from Merck, India. The commercial titanium dioxide (99.5%, Degussa P25 with 85 wt% anatase and 15 wt% rutile; average

particle size 25 nm, specific surface area 66.5 m²/g) and sodium hydroxide (98.5%) were supplied by Sigma Aldrich, USA. Nafion (perfluoro sulfonic acid-PTFE copolymer, 5% w/w in water solution) was purchased from Alfa Aesar, USA. Ethyl alcohol (99%) was purchased from Hayman group Ltd., Witham, England. All the chemicals were of analytical grade and used as received without any further purification. The water used in all preparations was distilled by a Millipore water purification unit (Elix-3, USA).

Ag-doped TiO₂ synthesis

Ag-doped TiO₂ was synthesized using AgNO₃ as Ag-metal precursor and AA as a reducing agent. 1 g of TiO₂ was added to 200 mL of AgNO₃ aqueous solution containing 1.5 g of Ag⁺ per 100 g TiO₂ (1.5% Ag (w/w)) under vigorous stirring (500 rpm for 60 min at room temperature). The pH of the suspension was adjusted to 7 using 0.1 M NH₄OH to achieve better adsorption of Ag⁺ onto TiO₂ support [43]. After reaching Ag⁺ sorption equilibration, the solution was centrifuged at 4500 rpm for 20 min to separate Ag⁺/TiO₂ from water. Ag⁺/TiO₂ residue was then dispersed in 25 mM L-ascorbic acid aqueous solution and stirred for 30 min. Upon the addition of ascorbic acid, a change in the color of the suspension from milky white to brown was observed which indicates the reduction of Ag⁺ to Ag⁰. The residue was then filtered out using a 0.2 µm membrane filter and washed four times with ethanol-water mixture and with DI water to remove loosely bound and/or undoped Ag⁺ on TiO₂. Finally, the residue was dried in a hot air oven at 100 °C for 12 h. The synthesized photocatalyst codes along with their nominal and true loadings are provided in Table S1 of Supplementary Information.

Physicochemical characterization

X-ray diffraction (XRD) of the photocatalysts was recorded by X-ray diffractometer (Model: Smartlab, Make: Rigaku, Japan) using Cu Kα radiation (λ = 0.1578 nm) with a step size of 0.01° and step speed of 20°/min. The optical absorbance was measured using diffuse reflectance UV–Vis spectrophotometer (UV–Vis, Model: UV-2600, Make: Shimadzu, Japan) over a spectral range of 200–800 nm with BaSO₄ as a reference. The concentration of Ag in Ag/TiO₂ was determined by using atomic absorption spectroscopy (AAS, Model: AA240FS, Make: Varian, The USA). For this purpose, the Ag present in Ag/TiO₂ was digested into an aqua regia solution. The electron dispersive X-ray spectroscopy (EDX, Model: Sigma, Make: Zeiss, Germany) was used for the surface elemental analysis. The morphology and microstructure of the samples were investigated using field emission scanning electron microscopy (FESEM, Model: JSM-7610F, Make: JEOL, Japan) and atomic force microscopy (AFM, Model: 5500 series, Make: Agilent, The USA). Raman spectra were measured by laser micro-Raman spectrometer (Raman, Model: LabRAM HR-800, Make: Horiba Jobin Yvon, The USA) with a 532 nm Ar laser excitation. The particle size and morphology were determined using field emission transmission electron microscopy (FETEM, Model: JEM-2100F, Make: JEOL, Japan). The sample was prepared for FETEM by dispersing a small mass in acetone using sonication, drop-casted onto the carbon-coated copper

grid and dried in vacuum. The surface area was obtained using an N₂ sorption study (Model: Tristar II, Make: Micromeritics, The USA). The samples were degassed at 200 °C for 6 h under vacuum to remove moisture and/or volatile compounds. Fourier transform infrared spectroscopy (FTIR, Model: IR Affinity 1, Make: Shimadzu, Japan) was performed to determine the pertinent functional groups of the specimens. Photoluminescence (PL, Model: Horiba Jobin Yvon, Make: Fluoromax-4, Japan) spectra were recorded under 280 nm excitation-emission spectra. To determine the oxidation state and surface composition of the prepared catalyst (Ag/TiO₂), X-ray photoelectron spectroscopy (XPS, Thermo Fisher Scientific Pvt. Ltd., The UK) analysis were carried out by Twin anode Non Monocromated ESCALAB Xi + instrument. Samples were prepared by dispersing the catalytic powders on the carbon tape. The working pressure of argon, pass energy and electron take-off angle was ~3 × 10^{−9} mBar, 20 eV and 55°, respectively. The full width at the half maximum of Au 4f_{7/2} was 0.8 eV (The reference XPS spectra of Au is shown in Fig. S1). Electron paramagnetic resonance (EPR, JES-FA200, USA) spectroscopy was employed with 1 mW microwave power and 9.42 GHz frequency.

Photoelectrochemical (PEC) characterization

Fabrication of 1.5Ag/TiO₂ thin film

1.5Ag/TiO₂ film was coated on indium tin oxide (ITO) substrate (surface resistivity 70–100 Ω sq^{−1}, Sigma Aldrich, India) using spin coating (Model: SpinNXXG-P1, Make: Apex Instruments, India). The substrates were cleaned by ultrasonication (Model: USC 200, Make: ANM industries, India) with acetone, ethanol, and DI water. A mass of 10 mg of 1.5Ag/TiO₂ was dispersed in 200 µL of a solution containing 0.5% (w/w) Nafion (binder) and isopropanol. The solution was ultra-sonicated for 2 h to obtain a homogenous mixture and then deposited onto the ITO substrate (1 cm × 1 cm) by spin coating at 2000 rpm for 10 s [44]. This specimen was then dried at room temperature to remove the solvent. The bare TiO₂ was also coated on the ITO substrate following a similar procedure. The thickness of the films was measured with stylus surface profilometer (Model: Dektak 150; Make: Veeco, USA) and it was found to be 310 and 340 nm for TiO₂ and 1.5Ag/TiO₂, respectively.

PEC measurements

Photocurrent properties of 1.5Ag/TiO₂ were determined using a precision Potentiostat/Galvanostat (Model: AUTOLAB 302N, Make: Metrohm Autolab B.V, Netherlands) with a three-electrode system. Photocatalyst coated ITO (1 cm²), platinum wire (1 mm diameter) and Ag/AgCl (3 M KCl) were used as working, counter and reference electrodes, respectively. All the PEC measurements were carried out in an undivided cell using 110 mL of 0.5 M Na₂SO₄ aqueous solution as an electrolyte. The photocurrents were measured by illuminating the photoanodes (TiO₂ and 1.5Ag/TiO₂) with a 500 W halogen lamp (450–550 nm). All the measurements were made at room temperature, atmospheric pressure and inside a dark chamber.

The electrical properties of the prepared thin films were determined by electrochemical impedance spectroscopy (EIS) using the same electrochemical work station. The EIS

measurements were carried out at 0 V vs. Ag/AgCl at the frequency from 0.1 to 10^5 Hz, and Mott-Schottky plots were taken at 10 kHz frequency from 0.0 to -1.4 V vs. Ag/AgCl in an aqueous solution. The Randell equivalent circuit was fitted to determine the resistance of the sample. Prior to each PEC experiment, the solution was purged with N_2 at a flow rate of 30 mL min^{-1} for 30 min to remove the dissolved oxygen.

Photocatalytic water splitting

The photocatalytic water splitting experiments were carried out in a closed gas circulation system with external irradiation of a quartz reactor with a capacity of 100 cm^3 . A 254 nm wavelength of UV light (Model: UV Lamp Pen-Ray, Make: Cole-Parmer India Pvt. Ltd, India) was used as a light source. The intensity of light was 4.40 mW cm^{-2} . The synthesized 1.5Ag/TiO₂ catalyst was suspended in 50 mL of solution. The suspension was sonicated (50 Hz) for 10 min before the reaction to avoid the particle aggregation to get a homogenous dispersion of catalyst. Prior to each photocatalytic experiment, the solution was degassed by flowing N_2 at 30 mL min^{-1} for 30 min. All the experiments were carried out in a black box to prevent interference from stray light. The evolved H_2 gas was analyzed at every 60 min by gas chromatography (Model: 7890A, Make: Agilent, The USA) equipped with a thermal conductivity detector and CP-molecular sieves column.

The apparent quantum yield (AQY) of H_2 evolution reaction was calculated using the following relation (Eq. (1)).

$$\text{AQY} = \frac{\text{Number of reacted electrons}}{\text{Number of incident photons}} \times 100\% \\ = \frac{2 \times \text{Number of } H_2 \text{ molecules evolved per hour}}{\text{Number of incident photons per hour}} \times 100\% \quad (1)$$

Results and discussion

Catalyst and electrode characterizations

The true loading of Ag and UV–Vis absorbance of Ag/TiO₂ samples with different wt% of Ag (0.5, 1, 1.5 and 2 wt%) were measured using AAS and UV–vis spectroscopy to determine the true loading of Ag (Table S1), bandgap (Table S2) and absorbance (Fig. S2). The analysis showed that the actual Ag concentration was 1.1% for 1.5% nominal concentration. Among various loadings tested, 1.5 wt% Ag/TiO₂ showed the highest absorbance in the visible range. Therefore, this photocatalyst was used for further study.

Spectroscopic analysis

UV–vis spectroscopy. UV–Vis diffuse reflectance spectra of bare TiO₂ and 1.5Ag/TiO₂ are shown in Fig. 1a. The pure TiO₂ spectrum showed an intense absorption in the UV region. The absorption spectrum of 1.5Ag/TiO₂ showed a distinct peak at 450 nm wavelength compared to that of bare TiO₂. The absorption of visible light by 1.5Ag/TiO₂ could be attributed to the SPR effect which is induced by the spatial confinement of electrons in Ag⁰ particles on the surface of TiO₂ [9]. The peak

broadening was observed towards a higher wavelength which could be due to the high dispersion of Ag on the TiO₂.

The optical bandgap energy (E_g) of the prepared samples was calculated using the Tauc plot (Eq. (2)).

$$\alpha h\nu = A (h\nu - E_g)^\eta \quad (2)$$

where α , $h\nu$, and A are the absorption coefficient, the energy of the incident photon and the proportionality constant, respectively. η is a constant which depends on the nature of transition, ($\eta = 2$ for the indirect bandgap of TiO₂) [45]. The bandgap was measured by extrapolation as shown in Fig. 1b (dashed lines). The bandgap energy of pure TiO₂ and 1.5Ag/TiO₂ were found to be around 3.0 and 2.5 eV, respectively. The decrease in the bandgap energy of Ag/TiO₂ was due to localized surface plasmon resonance (SPR) which in turn could be attributed to the doping of the noble metal Ag. In addition, the Ag⁺ ions substitute Ti⁴⁺ ions in the TiO₂ matrix and form electronic energy levels with a lower energy bandgap.

Raman spectra. The Raman vibrational spectroscopy of TiO₂ and 1.5Ag/TiO₂ could provide important insights into the phase composition, surface structure [46]. As evidenced in Fig. 1c, the sample exhibited a characteristic of the TiO₂ structure. TiO₂ contains both anatase and rutile phases and has a tetragonal symmetry with six ($A_{1g} + 2B_{1g} + 3E_g$) and four ($A_{1g} + B_{1g} + B_{2g} + E_g$) Raman active modes. The three distinct peaks at 397 (B_{1g}), 513 (B_{1g}) and 638 cm^{-1} (E_g) were observed corresponding to the anatase phase [47]. The peak corresponding to the rutile phase (E_g) appeared at around 448 cm^{-1} [20,48]. The intensity of the peaks for 1.5Ag/TiO₂ specimen was decreased significantly which could be attributed to the presence of adsorbed biomolecules [49].

Photoluminescence (PL) spectroscopy. Photoluminescence (PL) spectroscopy is a nondestructive technique to investigate the separation and recombination of photo-excited e^-/h^+ pairs in semiconductors and to explore the efficiency of the charge carrier trapping [50]. The excitation wavelength of 280 nm was used to collect the PL spectra of TiO₂ and 1.5Ag/TiO₂ semiconductors. As shown in Fig. 1d, both TiO₂ and 1.5Ag/TiO₂ showed similar PL spectra. The peak at 398 nm corresponds to the recombination, and the peaks in the range from 420 to 500 nm correspond to localized surface states of the valance band of TiO₂ [51,52]. The PL intensity of 1.5Ag/TiO₂ was lower compared to that of bare TiO₂ signifying a lower recombination rate of charge carriers.

FTIR spectra. FTIR spectra of TiO₂ and 1.5Ag/TiO₂ were recorded in the wavelength range of 4000–400 cm^{-1} , and are shown in Fig. 1e. The strong and broadband peak at around 2900–3600 cm^{-1} could be attributed to the stretching vibration of the hydroxyl group (O–H bending) of the TiO₂ surface. The peak at 1640–1630 cm^{-1} could be assigned to the bending frequency of water or Ti–OH, indicating the physisorbed OH groups or water molecules [53]. The bands at 650 and 1400 cm^{-1} were assigned to the lattice vibration of the Ti–O–Ti stretching mode [54]. The characteristic peaks at 800–400 cm^{-1} were assigned to the vibration of Ti–O–O. Due to the doping, a slight shift of peaks at around 660–700 cm^{-1}

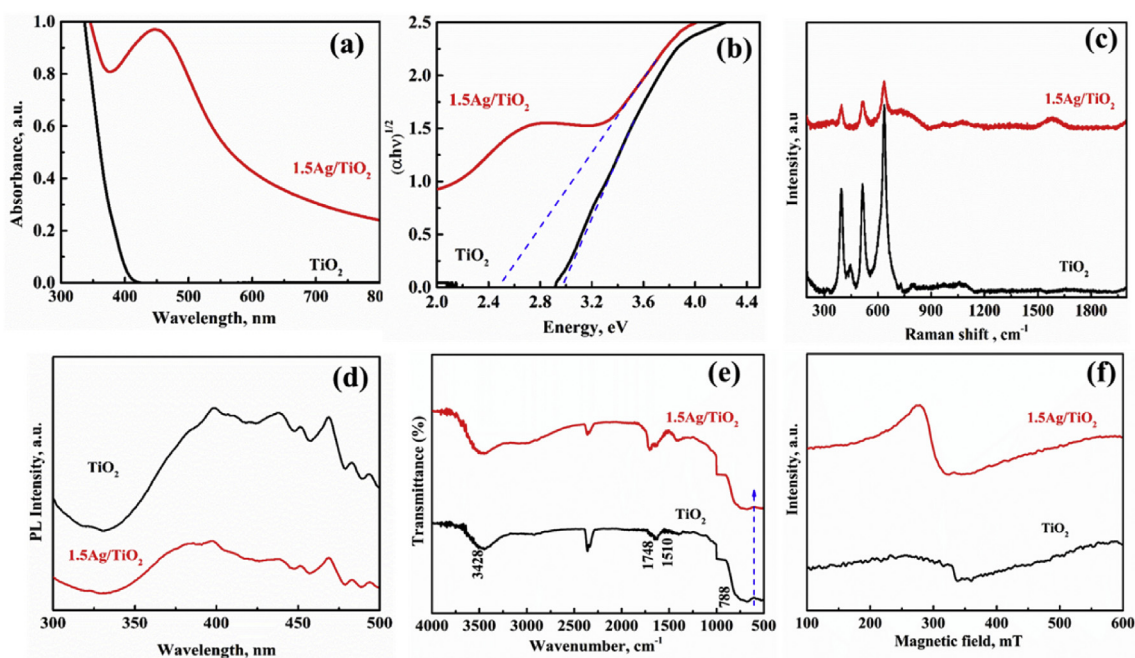


Fig. 1 – (a) UV–visible spectra, b) Tauc plot, c) Raman spectra, d) Photoluminescence spectra, e) FTIR spectra and f) EPR spectra of TiO_2 and $1.5\text{Ag}/\text{TiO}_2$.

was observed, indicating the asymmetrical vibration of $\text{Ti}-\text{Ag}-\text{O}$.

EPR spectroscopy. The EPR analysis was carried out under ambient conditions to detect oxygen vacancies and Ti^{3+} . As shown in Fig. 1f, TiO_2 showed a negligible EPR signal peak, while $1.5\text{Ag}/\text{TiO}_2$ exhibited a very intense EPR signal with g-factor of ~ 2.03 and ~ 1.96 . The peaks at ~ 310 mT and ~ 337 mT with respective above g-factors correspond to the formation of superoxide radicals and Ti^{3+} . These could be assigned to the unpaired electrons trapped on surface oxygen vacancies and the presence of paramagnetic Ti^{3+} species at the TiO_2 surface,

respectively [20,55]. Generally, metallic Ag could transfer e^- to react with Ti^{4+} or O_2 molecule to form Ti^{3+} and O_2^- [43,56]. Ti^{3+} act as electron traps and thus decrease the recombination rate, while oxygen vacancies help the adsorption of water molecules.

Structural analysis

X-ray diffractograms. The crystal phase and crystallite size of both bare TiO_2 and $1.5\text{Ag}/\text{TiO}_2$ were determined using XRD (Fig. 2 and Table 1). The peaks corresponding to the characteristic anatase (JCPDS 78–2486) and rutile (JCPDS 21–1276) phases were observed for both TiO_2 and $1.5\text{Ag}/\text{TiO}_2$ samples [57]. However, the diffraction peaks of Ag or Ag_2O were not observed for $1.5\text{Ag}/\text{TiO}_2$ which could be attributed to its higher dispersion on TiO_2 and/or lower concentration [58]. Generally, the XRD peaks corresponding to Ag appear at 38.1° and 64.5° when the Ag doping is more than 2 wt% [50]. The slight shift in the major peak of TiO_2 (25.40°) towards a lower diffraction angle of 25.37° (shown in the inset of Fig. 2a) indicates the induction of Ag^+ ions in the Ti^{4+} lattice [41].

The rutile phase content was calculated using the following equation (Eq. (3)).

$$\text{Rutile phase content (\%)} = \frac{A_R}{0.884 (A_A + A_R)} \quad (3)$$

where A_A and A_R represent the integrated intensities of the diffraction peaks from the (101) and (110) planes from the anatase and rutile phases, respectively. The anatase phase of TiO_2 was slightly decreased upon Ag-doping, as shown in Table 1 which is in-line with the literature reports [59]. Moreover, lattice strain was also slightly decreased (Table 1) with Ag-doping which may be due to the induction of Ag^+ into TiO_2 lattice forming $\text{Ti}-\text{Ag}-\text{O}$ [60]. In general, the decrease in the lattice strain and increase in the crystallite size of the

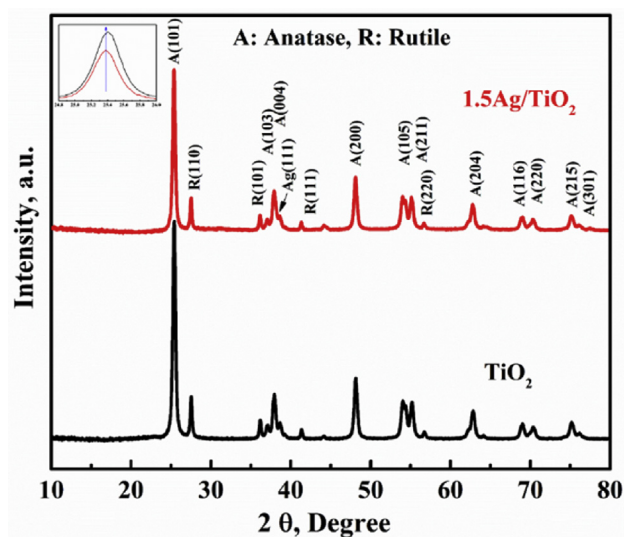


Fig. 2 – XRD patterns of TiO_2 and $1.5\text{Ag}/\text{TiO}_2$. (The shift in the 2θ can be seen in the inset).

Table 1 – Crystalline phase, average crystallite size, and lattice strain.

Photocatalysts	Crystalline phase, %		Crystallite size, nm		Lattice strain, ϵ
	Anatase	Rutile	Anatase	Rutile	
TiO ₂	79.23	20.77	21.95	31.17	0.0074
1.5Ag/TiO ₂	75.98	24.02	23.03	32.67	0.0072

anatase phase lowers the activation energy of the phase [50,61]. The larger ionic radius of Ag⁺ (126 Å) compared to Ti⁴⁺ (68 Å) allows only a small portion of the Ag⁺ going into the TiO₂ lattice by replacing the Ti⁴⁺ ions. The concentration of Ag⁺ in TiO₂ lattice is quantified using XPS analysis.

The average crystallite size of the photocatalysts was measured from the highest intensity peak using Scherrer's formula (Eq. (4)).

$$t = \frac{0.9\lambda}{\beta \cos \theta} \quad (4)$$

where λ is the X-ray wavelength, β is the full width at half maximum, and θ is the Bragg angle. The crystallite size of TiO₂ (Table 1) was increased upon Ag-doping which could be due to the induction of Ag into TiO₂ lattice.

X-ray photoelectron spectroscopy (XPS) analysis. The XPS analysis was employed to understand the chemical state of Ag

dopant and the interaction of Ag with TiO₂ lattice (Fig. 3). The spectra of Ag, O, and Ti were obtained by fitting of raw data (survey spectrum in Fig. 3a) with Gauss-Lorentz curve after subtraction of Shirley background using XPSPEAK4.1 software [62].

The high-resolution spectra of Ag 3d are shown in Fig. 3b. The same degree of shift in the binding energy was observed for both Ag 3d_{5/2} and Ag 3d_{3/2}. Therefore, we focused on the shift of Ag 3d_{5/2}. After the fitting, the binding energies (BEs) of Ag 3d_{5/2} were found to be 368.06 and 366.45 eV with an area percentage of 89.03 and 10.97%, respectively. The Ag 3d_{5/2} peak that was appeared at 368.06 eV with higher area percentage (89.03%) signifies Ag⁰ [63] While the peak appeared at lower binding energy, 366.45 eV (negative shift), signifies the possible electronic interaction between metal and support by charge transfer at the metal-support interface (electron transfer from TiO₂ to metallic Ag), forming Ti–Ag–O phase [64]. The Ag present in the Ti–Ag–O structure can be more

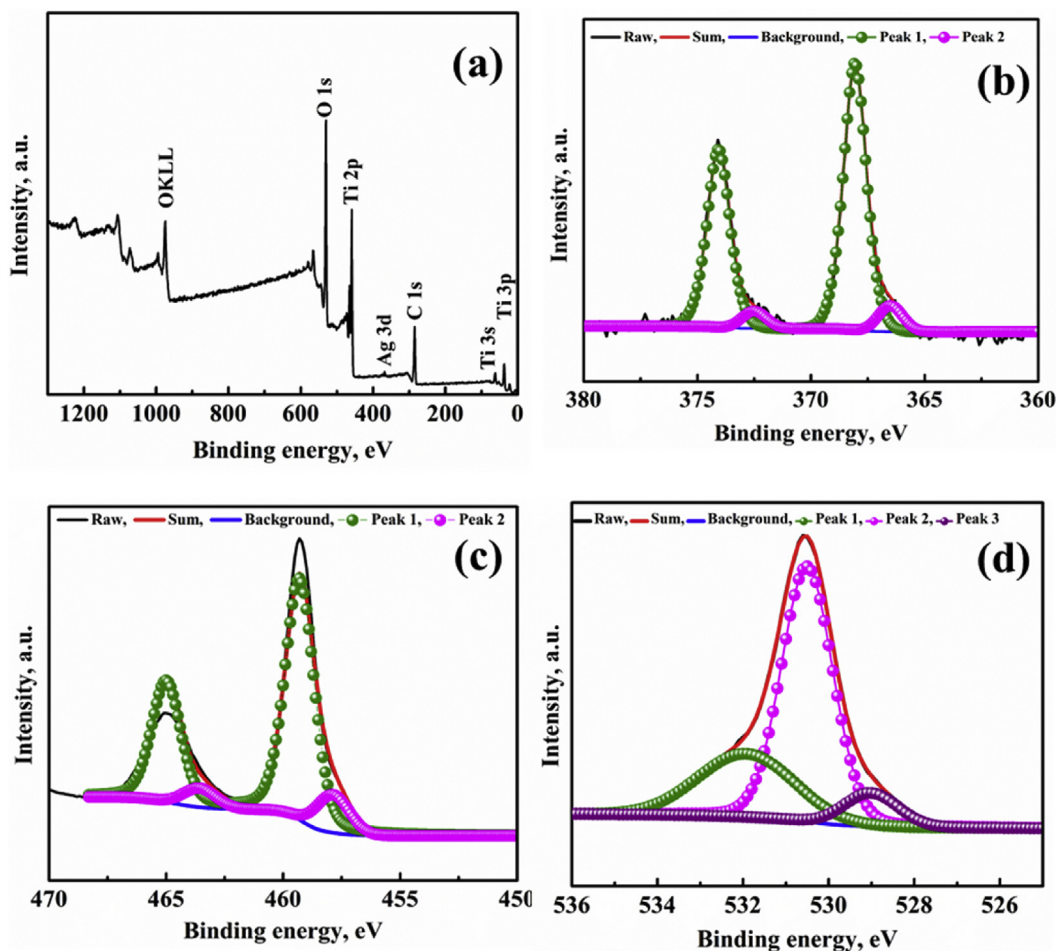


Fig. 3 – The XPS analysis: (a) Survey spectrum of 1.5Ag/TiO₂; the high resolution spectra of (b) Ag 3d, (c) Ti 2p and (d) O 1s.

negative than Ag^0 and can cause a negative shift in the binding energy due to the creation of an interface dipole ($\text{Ag}/\text{TiO}_2/\text{Ti}$) or attraction of free electrons from Ti to the Ag surface [65–69].

The high-resolution spectra of Ti 2p (Fig. 3c) show a higher peak area of $\text{Ti}^{3+} 2\text{P}_{3/2}$ at 459.29 eV as compared to that of $\text{Ti}^{4+} 2\text{P}_{3/2}$ at 457.86 eV which represents the removal of oxygen from the lattice of TiO_2 . The Ag-doping increased and decreased the areas of Ti^{3+} and Ti^{4+} peaks, respectively, indicating that either formation of Ti_2O_3 or mixed oxide structures of Ag and Ti^{3+} which was in-line with the corresponding Ag 3d spectra [70]. In principle, the creation of unpaired electrons or Ti^{3+} centers due to the oxygen vacancies affects the electron-hole recombination process in photocatalysts, causing a change in chemical rates that depend on charge transfer [71].

The formation of an oxygen vacancy of TiO_2 upon Ag-doping can be confirmed by the high-resolution spectra of O 1s. The O 1s spectra (Fig. 3d) shows three peaks at 529.01 and 530.25 and 531.92 eV which can be attributed to Ti–O, O–H and O-atoms in the vicinity of oxygen vacancies, respectively. High oxygen vacancies improve the charge transfer at the interface, which in turn improves the overall photocatalytic performance [72].

The increase in the C 1s atomic percentage from 21.27% (TiO_2) to 37.3% (1.5Ag/ TiO_2) (Table 2), could be due to the presence of AA used during the catalyst preparation.

Morphology studies

The surface morphologies of TiO_2 and 1.5Ag/ TiO_2 were captured using FESEM micrographs. The TiO_2 particles possessed nearly homogenous spherical shape and size (Fig. 4a and b). The Ag^0 was not visible as separate particles indicating that the dopant was uniformly distributed on TiO_2 . It also supports the X-ray diffraction pattern obtained for 1.5Ag/ TiO_2 . The EDX analysis was used to confirm the Ag presence in TiO_2 . The atomic weight percentages of Ag, O, and Ti elements in 1.5Ag/ TiO_2 were found to be 1.0 ± 0.2 , 44.2 ± 0.8 , and $54.8 \pm 0.7\%$, respectively (Fig. 4c). The elemental mapping is shown in Fig. 4d–g. It can be seen that Ag is homogeneously distributed throughout the surface of TiO_2 .

The FETEM micrographs were recorded to study the particle size and high-resolution morphology of 1.5Ag/ TiO_2 . The particle size distribution was in the range of 17–80 nm. The particles were slightly agglomerated as shown in the morphology (Fig. 4h). Further, localized doping of Ag on the surface of TiO_2 was observed from the HRTEM micrographs (Fig. 4i). The average d-spacing values of 0.35 nm correspond to the [101] plane of anatase TiO_2 and that of 0.24 nm corresponds to Ag [111] plane which shows a similar trend with XRD [20,73]. SAED pattern of 1.5Ag/ TiO_2 (Fig. 4j) represents the polycrystalline diffraction rings. The (hkl) planes ((101), (004) and (200)) of the anatase phase obtained from the SAED

patterns are also shown in Fig. 4. Overall, from FESEM, EDX and FETEM information, it was confirmed that Ag doping was uniform on the TiO_2 matrix.

PEC properties

The photoresponse of the prepared films was recorded by measuring the photocurrent density as a function of the applied potential (I–V characteristics) from –0.4 to 1.2 V vs. Ag/AgCl (3 M KCl) both in the dark and in the presence of light (Fig. 5a). At higher potentials, water oxidation took place and current density (CD) was increased significantly. 1.5Ag/ TiO_2 photoanode showed a higher CD as compared to that with TiO_2 . The major reasons for the higher CD with 1.5Ag/ TiO_2 are better electron-hole separation and SPR effect, as observed from PL and UV–Vis spectroscopies, respectively.

The chronoamperometric measurements were carried out by chopping the light in every 20 s to investigate the photo-sensitivity of the film (Fig. 5b). Upon illumination, a sharp increase in the anodic photocurrent was observed with 1.5Ag/ TiO_2 as compared to that with TiO_2 . 1.5Ag/ TiO_2 showed a higher transient photocurrent density of $1.16 \mu\text{A cm}^{-2}$ than that from TiO_2 ($0.33 \mu\text{A cm}^{-2}$) at 0.2 V vs. Ag/AgCl. This indicates the improvement of interfacial charge transfer between photoanode and electrolyte as well as an efficient charge carrier separation in metal and semiconductor interface [74].

The EIS spectra were recorded to study the charge transfer resistance (R_{ct}) at the interface of the photoelectrode and electrolyte solution [75]. The Nyquist plots obtained with TiO_2 and 1.5Ag/ TiO_2 photoanodes under dark and visible light illumination are shown in Fig. 5c. A smaller arc radius corresponds to an efficient charge transfer with a smaller charge transfer resistance at the semiconductor-electrolyte interface [76]. The radius of the arc with 1.5Ag/ TiO_2 was much smaller than that with TiO_2 . The R_{ct} in parallel with double-layer capacitance (C_{dl}) at the interface was estimated by the fitting of semi-arcs. The values of R_{ct} under dark for TiO_2 and 1.5Ag/ TiO_2 photoanodes were found to be 26×10^5 , and $4.2 \times 10^5 \Omega$ and those under light were 7.3×10^5 and $2.5 \times 10^5 \Omega$, respectively. The values indicate that the resistance was lower with Ag-doped TiO_2 as compared to that with bare TiO_2 , under both dark and illuminated conditions. The lower R_{ct} endorses a faster charge transfer and better electron mobility in the Ag-doped photocatalyst [20]. These results are in line with PL spectroscopy where lower recombination of photogenerated charge carriers was observed with 1.5Ag/ TiO_2 (Fig. 1d).

In order to determine the flat band potentials (E_{fb}), Mott-Schottky (MS) plot ($1/C^2$ versus potential) at a fixed frequency of 10 kHz in 0.5 M aqueous Na_2SO_4 electrolyte solution was plotted for both TiO_2 and 1.5Ag/ TiO_2 electrode systems (Fig. 5d). The capacitance-potential relation can be represented by the following equation (Eq. (5)) [77]:

$$\frac{1}{C^2} = \frac{2}{N_d \epsilon \epsilon_0 e A^2} \left[\left(E - E_{fb} - \frac{kT}{e} \right) \right] \quad (5)$$

where C is the interfacial capacitance, N_d is the charge carrier density, ϵ is the dielectric constant, ϵ_0 is the permittivity of the vacuum, and A is the active surface area. E_{fb} was measured by extrapolating a line from the linear relationship of $\frac{1}{C^2}$ versus E.

Table 2 – Atomic percentage (%) of Ti, O, C, Ag in TiO_2 and 1.5Ag/ TiO_2 from XPS analysis.

Photocatalysts	Ti 2p	O 1s	C 1s	Ag 3d ₃	Ag 3d ₅
TiO_2	25.58	52.57	21.27	–	–
1.5Ag/ TiO_2	16.87	45.66	37.3	0.04	0.06

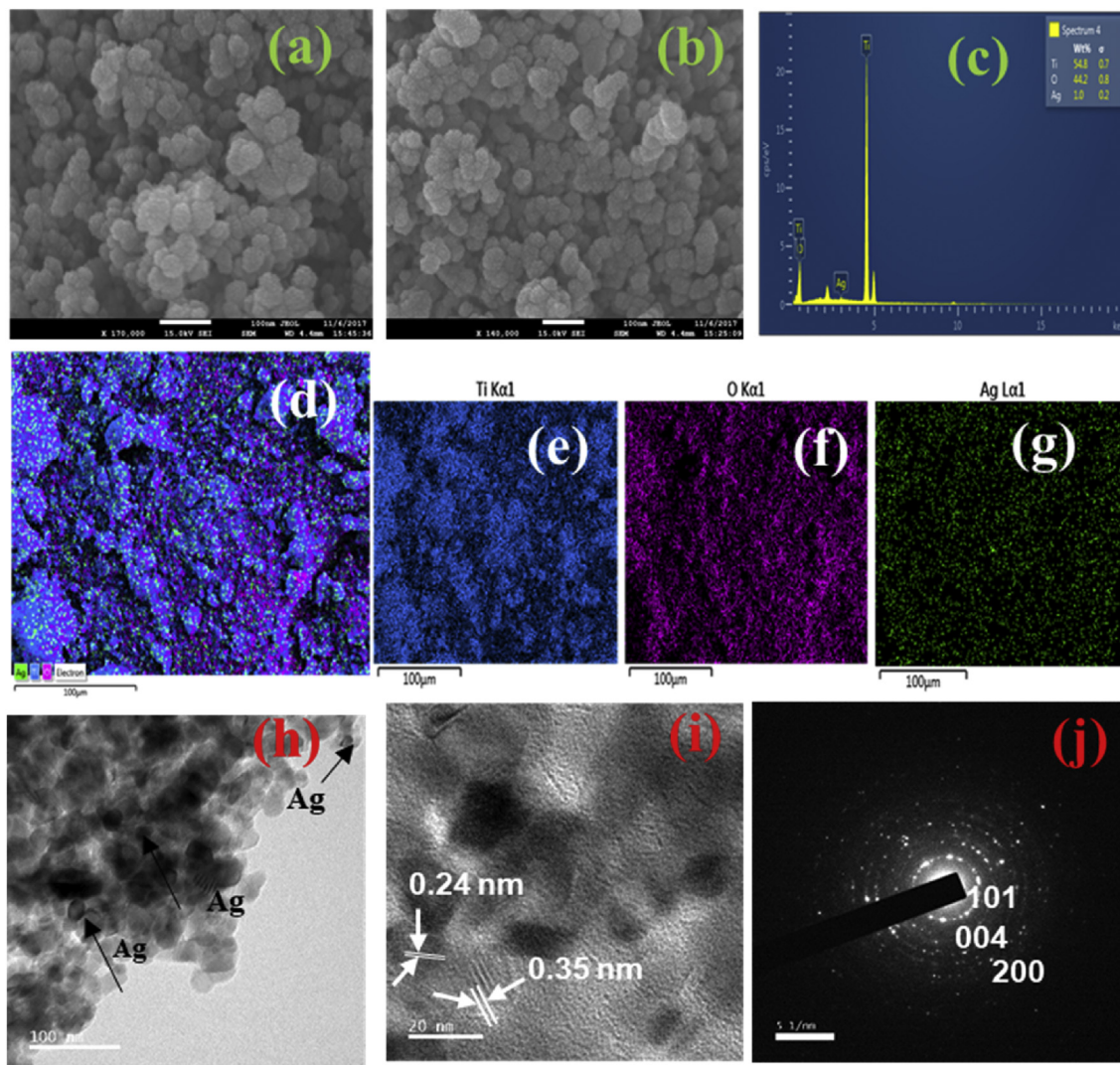


Fig. 4 – FESEM images of (a) bare TiO_2 and (b) $1.5\text{Ag}/\text{TiO}_2$, (c) EDX analysis of $1.5\text{Ag}/\text{TiO}_2$, (d) Elemental mapping of Ti, O and Ag, (e) Dispersivity of Ti, (f) Dispersivity of O, (g) Dispersivity of Ag, (h) FETEM image of $1.5\text{Ag}/\text{TiO}_2$, (i) HRTEM image of $1.5\text{Ag}/\text{TiO}_2$ and (j) SAED pattern of $1.5\text{Ag}/\text{TiO}_2$.

The value of E_{fb} was found to be -1.34 and -0.95 V vs. Ag/AgCl (3 M KCl) with TiO_2 and $1.5\text{Ag}/\text{TiO}_2$, respectively. The corresponding E_{fb} values with respect to RHE (Reference Hydrogen Electrode) were -0.72 and -0.33 V (from the Nernst equation (Eq. (6)). For n-type semiconductors, the CB potential is often considered to be less than 0.1 V of its E_{fb} [78,79]. Therefore, the CBs of TiO_2 and $1.5\text{Ag}/\text{TiO}_2$ were calculated to be -0.82 and -0.43 V vs. RHE, respectively. The bandgaps of TiO_2 and $1.5\text{Ag}/\text{TiO}_2$ are 3.0 eV and 2.5 eV, respectively. The corresponding VB potential of TiO_2 and $1.5\text{Ag}/\text{TiO}_2$ were determined as 2.18 and 2.07 V, respectively ($\text{VB} = \text{CB} + \text{Bandgap}$) [76]. The doping of Ag pushes both the CB and VB edge of TiO_2 towards the potentials of water reduction and oxidation, respectively. Thus, a positive shift with $1.5\text{Ag}/\text{TiO}_2$ could facilitate the electron transfer by improving the electronic state distribution [80]. The N_d values as obtained from the slope of the graph were $1.08 \times 10^8 \text{ cm}^{-3}$ and $1.25 \times 10^8 \text{ cm}^{-3}$ with TiO_2 and $1.5\text{Ag}/\text{TiO}_2$, respectively. The increased charge carrier density and

reduced charge recombination upon Ag doping are the major contributing factors for the photocurrent enhancement of $1.5\text{Ag}/\text{TiO}_2$.

$$E_{\text{RHE}} = E_{\text{Ag}/\text{AgCl}} + 0.059\text{pH} + E_{\text{Ag}/\text{AgCl}}^0 \quad (6)$$

Photocatalytic water splitting

The control experiments with no photocatalyst (TiO_2 and $1.5\text{Ag}/\text{TiO}_2$) and/or light showed no formation of hydrogen.

Effect of catalyst concentration

The effect of the concentration of $1.5\text{Ag}/\text{TiO}_2$ and reaction time on the water splitting activity was studied in order to find a suitable concentration for efficient photon absorption (Fig. 6). The cumulative H_2 production was almost linearly increased with reaction time at all the catalyst concentrations (Fig. 6a). At lower times (i.e., <240 min), the H_2 production rate

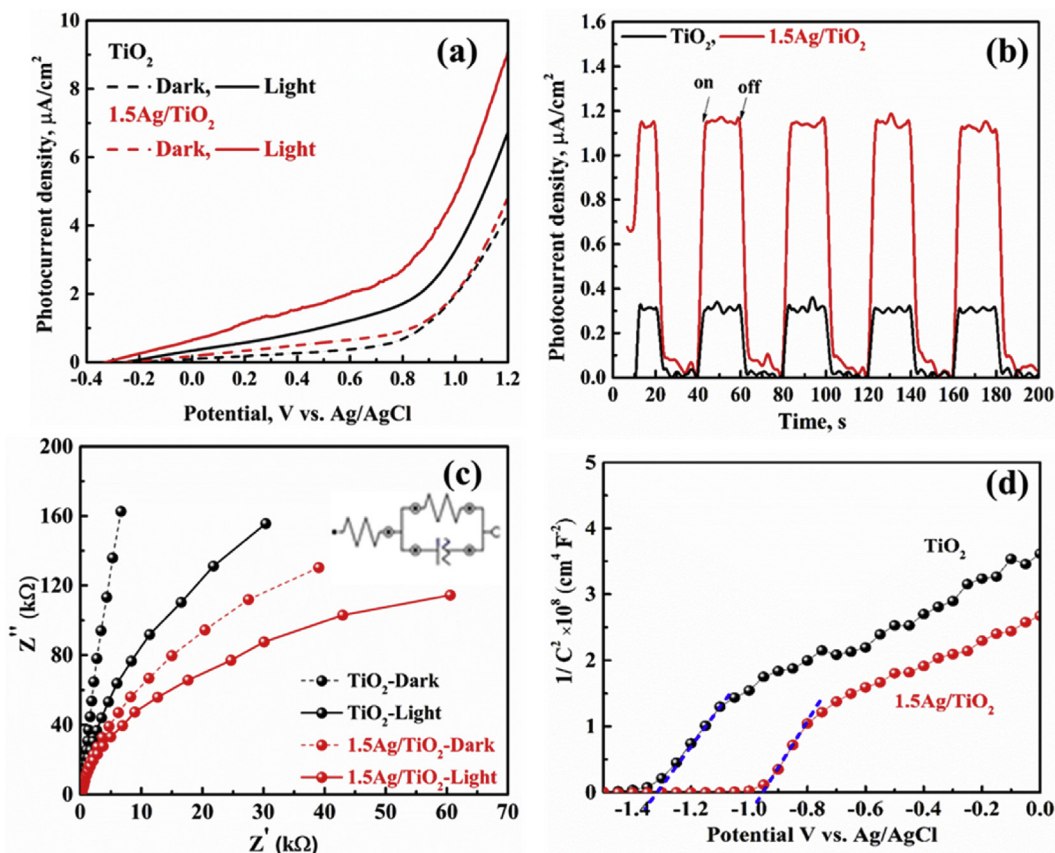


Fig. 5 – (a) Current density of the coated thin films as a function of potential vs. Ag/AgCl, (b) Chronoamperometric measurements (photocurrent response in dark and light as a function of time), (c) Nyquist plots, and (d) Mott-Schottky plots with TiO_2 and $1.5\text{Ag}/\text{TiO}_2$ electrode.

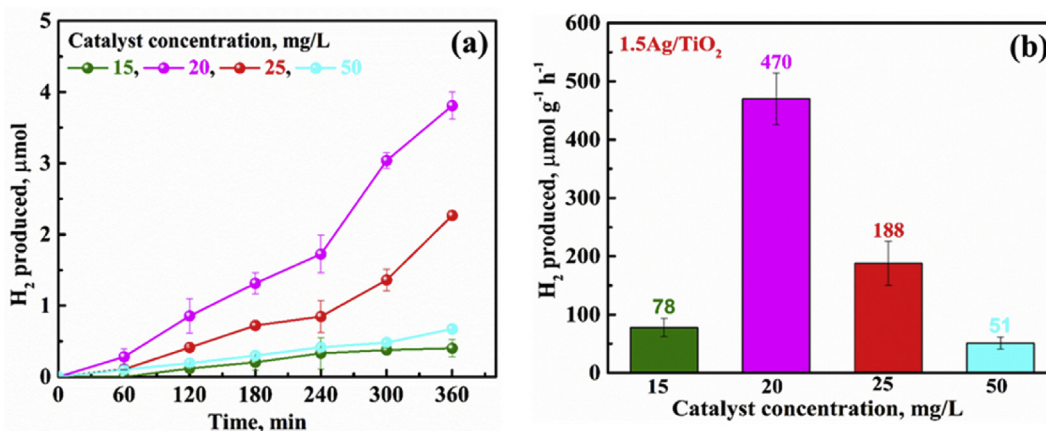


Fig. 6 – Effect of catalyst concentration on water splitting: (a) H_2 production (μmol) as a function of time (min) at various catalyst concentrations and (b) average H_2 production rate ($\mu\text{mol g}^{-1} \text{h}^{-1}$) as a function of catalyst concentration.

was observed to be lower. A sudden jump in H_2 production rate was observed at higher times (>240 min). This trend was observed only with the $1.5\text{Ag}/\text{TiO}_2$ but not with TiO_2 (Fig. 9a) photocatalyst. This could be attributed to the presence of small amounts of AA on the surface of the metal (Ag) which in turn lowers the effective contact of the reactant with the photocatalyst [81]. As the contact time increases, the AA layer dissolves into the solution, the Ag surface exposes to the

reactant and activity increases. The presence of AA was confirmed by the characterizations such as RAMAN spectroscopy (Fig. 1c). The H_2 evolution rate was first increased with the catalyst loading from 15 to 20 mg L^{-1} and then decreased up to 50 mg L^{-1} (Fig. 6a and b). The highest photocatalytic activity of $470 \mu\text{mol g}^{-1} \text{h}^{-1}$ was achieved with a photocatalyst concentration of 20 mg L^{-1} in the absence of SR. The main reason for the decrease in activity at higher

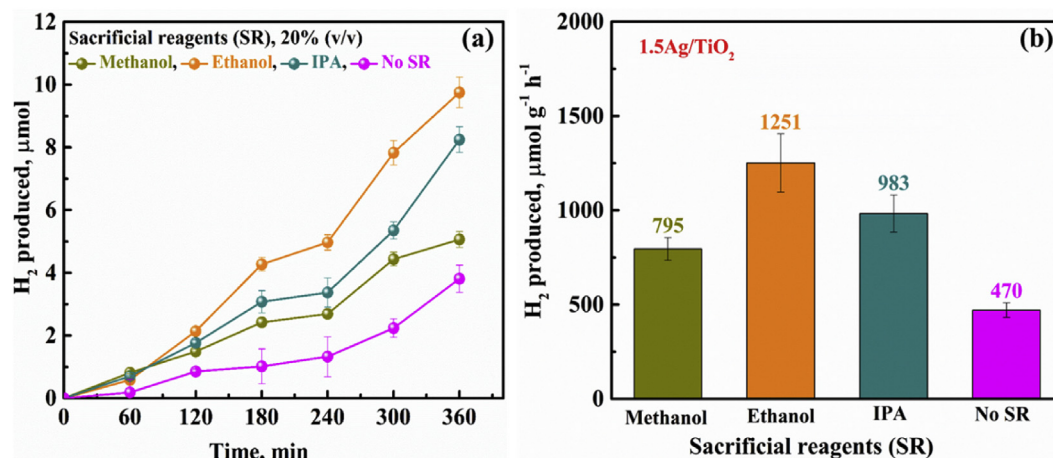


Fig. 7 – Effect of alcohol as SR on water splitting: (a) H₂ production (μmol) as a function of time in the presence of various alcohols and (b) Average H₂ production rate (μmol g⁻¹ h⁻¹).

concentrations was the reduction in penetration depths of light into the suspension. An excess catalyst loading may hinder radiation absorption and in some cases, it could promote the recombination of photogenerated e⁻/h⁺ [82]. Overall, H₂ evolution was decreased in highly turbid suspensions due to particle agglomeration and radiation shadowing phenomena. Therefore, a catalyst concentration of 20 mg L⁻¹ was used in further study.

Effect of sacrificial reagents (SRs)

Alcohols as SRs. The influence of different alcohols as SRs on water splitting reaction was studied using 20% (v/v) methanol, ethanol or isopropanol (IPA) (Fig. 7a & b). H₂ evolution rate was higher with ethanol (1251 μmol g⁻¹ h⁻¹) as compared to that with methanol (795 μmol g⁻¹ h⁻¹) and IPA (983 μmol g⁻¹ h⁻¹) under otherwise identical conditions. According to Strataki et al. [83], ethanol gives the highest photocatalytic activity compared to methanol at a lower concentration of alcohol (<5% v/v). Oxidation of adsorbed water molecules by hydroxyl radicals is predominant at a low alcohol concentration [84]. At a

higher concentration (>80% v/v), direct oxidation dominates where methanol traps the holes more effectively to give a higher H₂ yield. Other factors, such as adsorption of intermediates and solvent polarity, favor a higher H₂ production in the case of ethanol over methanol [85]. In general, ethanol strongly adsorbs onto TiO₂ than on the metal surface due to a stronger heat of adsorption of oxide. At the interface between metal and TiO₂, the adsorbed ethanol is dissociated into surface ethoxide and hydroxyl group where the ethoxide species act as hole traps and hydroxyl species could easily be reduced by electrons from the metal particles [86].

Na₂S and Na₂SO₃ as SR. The influence of the mixture of 0.1N Na₂S and 0.1N Na₂SO₃ as SRs were also examined with different catalyst concentrations (Fig. 8). A similar trend as without SRs was observed, giving the highest H₂ evolution rate with 20 mg L⁻¹ of photocatalyst concentration. The maximum H₂ evolution rate over 1.5Ag/TiO₂ (23496 μmol g⁻¹ h⁻¹) in the presence of SR was 50 times higher than that (470 μmol g⁻¹ h⁻¹) in its absence. Sulfur as a sacrificial agent is

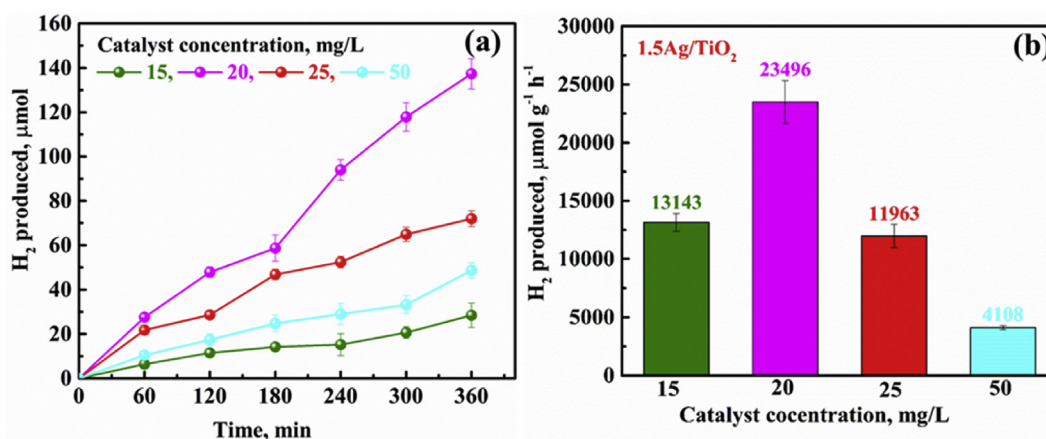


Fig. 8 – Effect of catalyst (1.5Ag/TiO₂) concentration in the presence of 0.1N Na₂S and 0.1N Na₂SO₃ as SRs: (a) H₂ production (μmol) as a function of time and (b) Average H₂ production rate (μmol g⁻¹ h⁻¹) with various catalyst concentrations.

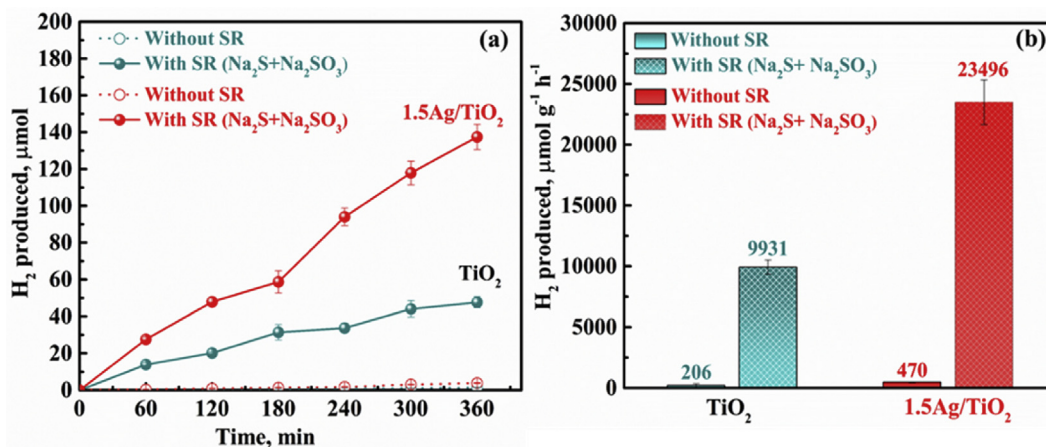
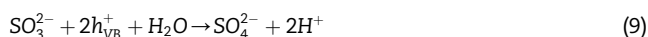
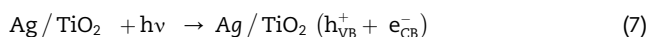


Fig. 9 – Photocatalytic water splitting over bare TiO₂ and 1.5Ag/TiO₂ in the presence and absence of 0.1N Na₂S and 0.1N Na₂SO₃ as SRs: (a) H₂ production (μmol) as a function of time and (b) Average H₂ production rate (μmol g⁻¹ h⁻¹).

effective in hole trapping from the valance band which is evident from the Fig. 8a and b. The advantages of using Na₂S and Na₂SO₃ as sacrificial reagents are lower energy consumption and easy oxidizability compared to alcohol.

The role of SRs in photocatalytic water splitting reaction can be explained with the help of the following reactions (Eqs. (7)–(12)) [87].



Photoexcitation of a semiconductor by light with energy more than bandgap could lead to the excitation of electrons to the CB, leaving the holes in the VB (Eq. (7)). The photo-generated electrons migrate to the surface of the catalyst where they undergo reactions with the adsorbed water molecules forming H₂ (Eq. (8)). Meanwhile, the holes could react with the sacrificial ions by oxidizing SO₃²⁻ to SO₄²⁻ and S₂²⁻ (Eqs. (9) and (10)). A mutual reaction between S₂²⁻ and SO₃²⁻ could yield a colorless compound, S₂O₃²⁻ (Eq. (11)). Furthermore, an excess S²⁻ ion is scavenged by SO₃²⁻ (Eq. (12)). To check the surface modification and/or poisoning of the catalyst due to sulfur-based sacrificial reagents, XRD (Fig. S3a) and EDX (Fig. S3b) analysis of the used photocatalyst were performed after 6 h of reaction. No significant change in the crystal structure and deposition of sulfur on the catalyst surface were observed.

Activity comparison of TiO₂ and 1.5Ag/TiO₂

H₂ production rates over bare TiO₂ and 1.5Ag/TiO₂ in pure water as well as in the presence of Na₂S (0.1 N) and Na₂SO₃

(0.1 N) as SRs were compared under UV light (Fig. 9). The rate of H₂ production over 1.5Ag/TiO₂ was higher, in both the presence and absence of SRs, (with SR: 23496 μmol g⁻¹ h⁻¹ in comparison to that over bare TiO₂ (9931 and 206 μmol g⁻¹ h⁻¹). The apparent quantum yields achieved with 1.5Ag/TiO₂ and bare TiO₂ were 19 and 9%, respectively, at 254 nm wavelength in the presence of SR. The higher activity is mainly attributed to the oxygen vacancies and the strong metal-support interaction of Ti–Ag–O species, as depicted from XPS analysis. Moreover, the homogenous distribution of Ag (EDX) and a narrow bandgap (UV–vis) led to a higher H₂ yield. In addition, the enhanced physicochemical properties such as high photosensitivity, efficient charge transfer, and better electronic states distribution, as inferred from PEC measurements, may be accountable to drive efficient water splitting. The presence of both anatase and rutile phases in the 1.5Ag/TiO₂ creates ‘catalyst hotspots’ because of rapid charge transfer dynamics which in-turn extends the lifetimes of e⁻/h⁺ pair [88]. Therefore, the synergistic effect of the mixed phases and oxygen vacancies may also contribute to a higher catalytic activity by enhancing the photogenerated electron/hole pair’s separation.

A comparison of water splitting performance of 1.5Ag/TiO₂ synthesized in the present work with that of literature reported catalysts is presented in Table S4. A higher H₂ rate was obtained in the present study as compared to that in the literature reports. A remarkable improvement of H₂ production using sulfur-based SR was observed due to the effective scavenging of holes. Moreover, the present work showed a higher H₂ yield at a lower concentration of methanol compared to previous studies. Jovic et al., 2014 achieved a higher H₂ yield (3700 μmol g⁻¹ h⁻¹) with 80% ethanol concentration. Whereas, in the present study, a yield of 1251 μmol g⁻¹ h⁻¹ was obtained with 20% ethanol under UV-excitation [89].

Reusability of 1.5Ag/TiO₂ photocatalyst

The reusability of catalyst was tested in the absence and presence of SRs in three consecutive test cycles (Fig. 10). The reaction mixture was purged with N₂ for 30 min after each 6 h

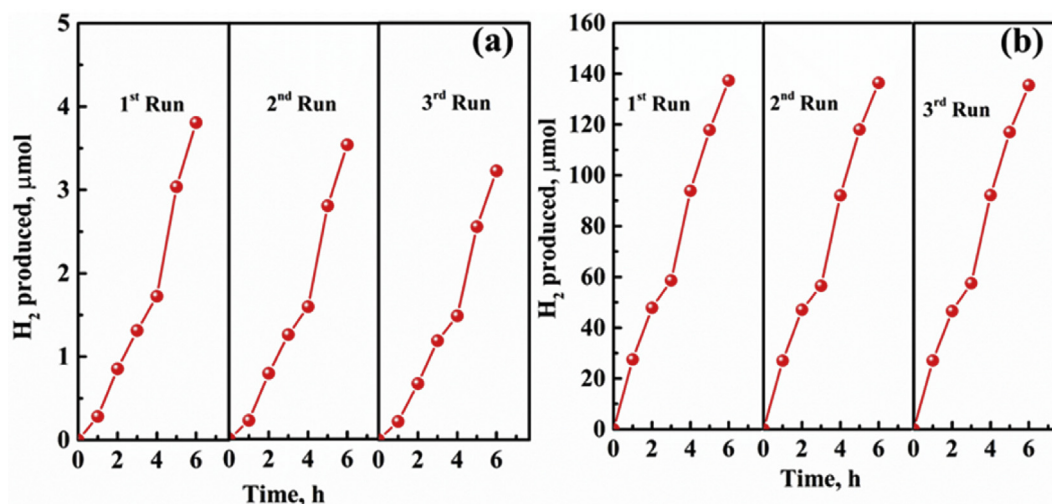


Fig. 10 – Recycle test for the photocatalytic water splitting over 1.5Ag/TiO₂ in both absence (a) and presence (b) of SR.

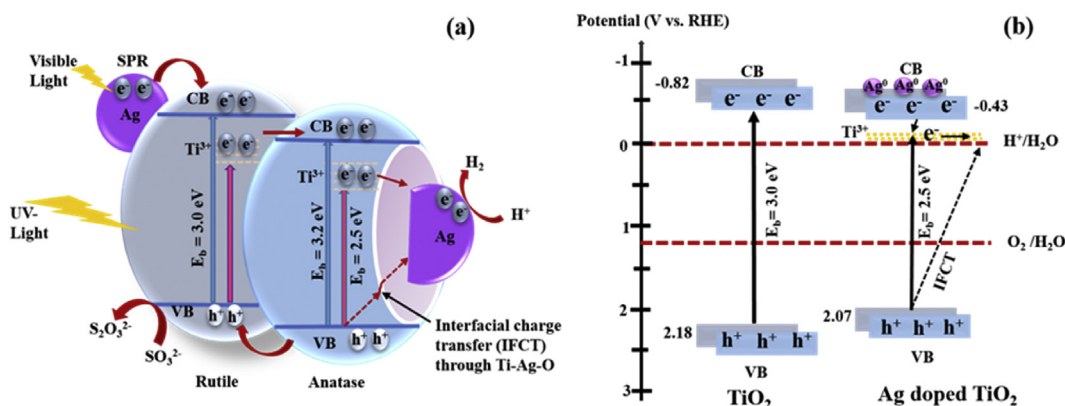


Fig. 11 – (a) Schematic representation of electrons generation, separation, and participation in water splitting over Ag-doped TiO₂ (Red arrows indicate the flow of electrons and holes), (b) Schematic diagram of the CB and VB position of TiO₂ and 1.5Ag/TiO₂ against RHE. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

reaction and the experiment was repeated. The activity of the photocatalyst in both the cases retained until the three cycles, offering the efficient reusability of the photocatalyst. In each cycle, the average rate of reaction was measured from the slope of the graphs which was found to be nearly a constant ($\sim 0.54 \mu\text{mol h}^{-1}$ in pure water and $\sim 22.8 \mu\text{mol h}^{-1}$ in the presence of SR). In both cases, a cumulative increase of H₂ generation with the duration of irradiation showed the consistent zero-order kinetics with no noticeable decay of the photocatalyst [90].

Mechanism of enhanced H₂ production from Ag/TiO₂

The excitation mechanism of electron-hole (e^-/h^+) pair in the mixed-phase (Ag⁰ and Ti–Ag–O species) on the TiO₂ surface is shown in Fig. 11. The plasmonic Ag⁰ absorbs visible light (SPR effect), generating hot e^- and transferring them to rutile-TiO₂ [91]. The hot electrons then transfer from the CB of rutile to the anatase because of the more negative CB level of rutile than anatase [92]. At the same time, the generated holes from the VB of anatase transfers to the VB of rutile-TiO₂ where the

holes further react with water or SR. Furthermore, the presence of Ti³⁺ and oxygen vacancies as revealed by XPS can effectively narrow down the bandgap of both rutile and anatase phases. The electrons on Ti³⁺ band transfer to the surface of silver species efficiently [93]. Another pathway of excited electrons from the VB to the surface of silver species is by interfacial charge transfer (IFCT). This pathway is particularly favorable in the presence of the Ti–Ag–O phase and accounts for enhanced activity [21]. The proposed mechanism helps in enhanced photocatalysis by reducing the recombination of photogenerated charge carriers through electron trapping at the metal-semiconductor interface.

Conclusions

The Ag/TiO₂ photocatalyst was successfully synthesized using chemical reduction with ascorbic acid. The doped catalyst has a distinct absorption peak and its bandgap reduced to 2.5 eV as compared to that of TiO₂ (3.0 eV). The EDX confirmed the

uniform distribution of Ag on the TiO₂ surface. PL and EIS demonstrated the efficient interfacial charge transfer from TiO₂ to Ag, possibly through the Ti–Ag–O bonds. The XPS analysis suggested the existence of oxygen vacancies in TiO₂ and 11% Ti–Ag–O phase. A higher transient photocurrent density of 1.16 $\mu\text{A cm}^{-2}$ was observed over 1.5Ag/TiO₂ as compared to that over TiO₂ (0.33 $\mu\text{A cm}^{-2}$), indicating an efficient charge carrier separation over 1.5Ag/TiO₂. The highest H₂ production rate of 23.5 mmol g^{−1} h^{−1}, with an apparent quantum yield of 19%, was obtained in the presence of 0.1 N Na₂S and 0.1 N Na₂SO₃ sacrificial reagent (SR) over 1.5Ag/TiO₂. The best H₂ evolution rate (1251 $\mu\text{mol g}^{-1}$ h^{−1}) was obtained with ethanol as compared to other alcohols tested as SRs in this study. This study provides a new understanding of the formation of mixed-phase containing metallic Ag and Ti–Ag–O which enhances the charge separation and results in the higher hydrogen production rates.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2019.11.127>.

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